

Fake Currency Detection Using Deep Learning

A project report submitted to

MALLA REDDY UNIVERSITY

In partial fulfilment of the requirements for the award of degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING (AI & ML)

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2023



MALLA REDDY UNIVERSITY

(Telangana State Private Universities Act No.13 of 2020 and G.O.Ms.No.14, Higher Education (UE) Department)

COLLEGE CERTIFICATE

This is to certify that this is the bonafide record of the application development entitled “**Fake Currency Detection Using Deep Learning**” submitted by **Avinash Parthan : 2011CS020027, Ayush Rathore : 2011CS020028, B. Pranitha : 2011CS020029, B. Sai Varun : 2011CS020030** of B. Tech III year II semester, Department of CSE (AI & ML) during the year 2022-23. The results embodied in this report have not been submitted to any other university or institute for the award of any degree or diploma.

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ACKNOWLEDGEMENT

We would like to express our gratitude to all those who extended their support and suggestions to come up with this project. Special Thanks to our mentor Dr.

S. Satyanarayana whose support, stimulating suggestions and encouragement helped us all time in the due course of project development.

We sincerely thank HOD Dr. Thayyaba Khatoon for her constant support and motivation all the time. Last but not the least our sincere appreciation goes to our families who has been tolerant understanding our moods and extending timely support.

ABSTRACT

Proposed a novel approach for identifying counterfeit banknotes using deep learning techniques. The proposed method involves pre-processing the images of banknotes and then using a Convolutional Neural Network (CNN) to extract features. The CNN is trained on a large dataset of genuine and counterfeit banknotes to learn the underlying patterns of genuine notes and distinguish them from fake ones. The experimental results demonstrate the effectiveness of the proposed method in detecting fake currency with high accuracy, which can be used to prevent economic losses and protect consumers from fraud. The proposed method has the potential to be implemented in automated systems to detect counterfeit banknotes in real-time.

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CHAPTER - 1

INTRODUCTION

Fake currency detection is a critical task that plays a significant role in maintaining the integrity of a country's currency. With the increasing amount of fake currency in circulation, there is a need for automated and efficient fake currency detection systems. Machine learning has emerged as a promising technique for fake currency detection due to its ability to analyze large amounts of data and identify patterns that can be used to distinguish between real and counterfeit currency.

1.1 Problem Statement

Counterfeit currency is a significant problem worldwide, leading to financial losses and undermining trust in monetary systems. Traditional methods of counterfeit detection are time-consuming and often rely on human expertise, which can be error-prone. There is a need for an automated and accurate solution to detect fake currency notes efficiently.

1.2 Objective of Project

The objective of the Fake Currency Detection using Deep Learning project is to develop a software system that can automatically detect counterfeit currency notes using deep learning techniques. The system aims to provide a reliable and efficient solution to identify fake currency and distinguish it from genuine currency notes.

1.3 Scope & Limitations of project

Scope

The project's scope includes the development of a deep learning-based software system capable of analyzing images of currency notes and classifying them as genuine or counterfeit. The system will be designed to handle different types of currency notes and variations in image quality.

Limitations of project

However, it should be noted that the system's accuracy may be influenced by the quality of input images and the complexity of counterfeit techniques employed

- ❖ Limited parameters are considered for the study.
- ❖ Only selected DL techniques are being used to ascertain the accuracy given by the proposed model...

1.4. Existing System

The existing systems for counterfeit currency detection primarily rely on manual inspection by experts, which can be time-consuming, subjective, and error-prone. Some existing methods use traditional computer vision techniques and feature extraction algorithms, but they may not be robust enough to handle sophisticated counterfeit techniques.

1.5. Proposed System

The proposed system utilizes deep learning techniques to automatically detect counterfeit currency notes. It involves preprocessing the input currency images, extracting relevant features using deep learning algorithms, training a model with a labeled dataset of genuine and counterfeit currency images, and classifying new images based on the trained model. The system aims to provide accurate and real-time detection results, enhancing the efficiency and reliability of counterfeit currency detection.

1.6. Modules

The Fake Currency Detection system can be divided into the following modules:

1. Image Input: This module allows users to input images of currency notes for analysis.
2. Preprocessing: The module preprocesses the input images to enhance their quality and extract relevant features.
3. Feature Extraction: This module extracts distinctive features from the preprocessed images using deep learning algorithms.
4. Training: The system trains a deep learning model using a labeled dataset of genuine and counterfeit currency images.
5. Classification: The module classifies the input images as genuine or counterfeit based on the trained model.
6. Confidence Level: This module provides a confidence level or probability score for the classification result.
7. Reporting: The system generates a report summarizing the results of the counterfeit currency detection process.

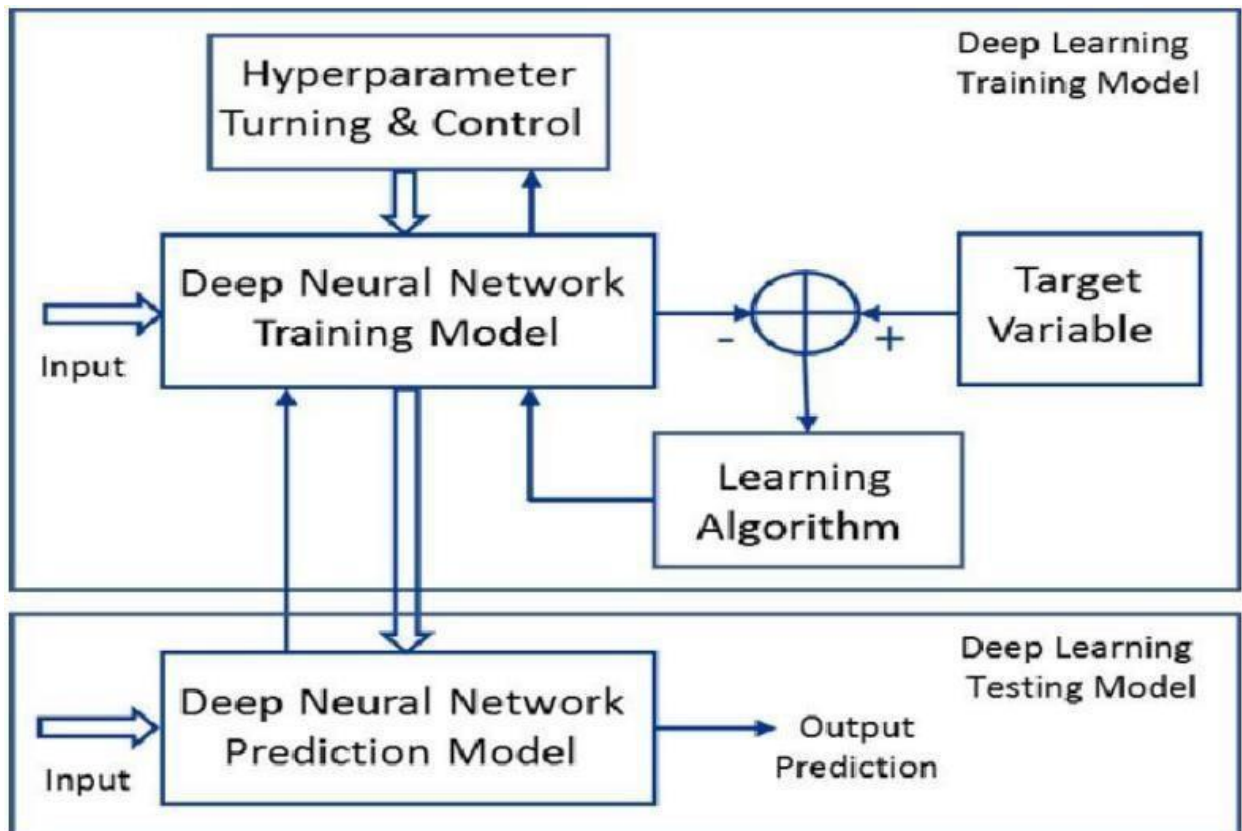
1.7. Architecture

The system architecture follows a layered approach. It includes the following components:

1. User Interface: Provides an interface for users to interact with the system, input currency images, and view detection results.
2. Preprocessing Layer: Handles the preprocessing of input currency images, including image enhancement and feature extraction.
3. Deep Learning Layer: Trains and utilizes deep learning models to classify the currency images.

4. Database Layer: Stores the labeled dataset, trained models, and detection results for future reference.
5. Reporting Layer: Generates reports summarizing the detection results and other relevant information.

Overall, the proposed system aims to automate counterfeit currency detection using deep learning, enhancing the accuracy, speed, and efficiency of the process while minimizing reliance on human expertise.



CHAPTER – 2

LITERATURE SURVEY

Automatic Counterfeit Banknote Recognition Using Convolutional Neural Networks by Bhowmik et al. (2017):

This paper proposes a deep learning-based approach using Convolutional Neural Networks (CNNs) for counterfeit banknote recognition. The authors trained a CNN model on a large dataset of genuine and counterfeit banknote images and achieved high accuracy in distinguishing between genuine and counterfeit notes.

Fake Currency Detection Based on Convolutional Neural Network by Jaiswal and Gupta (2018):

In this paper, the authors proposed a fake currency detection system based on a CNN model. They used a dataset of currency note images and employed various image processing techniques for preprocessing. The CNN model was trained to classify images as genuine or counterfeit with high accuracy.

Detection of Counterfeit Indian Currency Using Convolutional Neural Network by Saini et al. (2019):

The authors presented a counterfeit Indian currency detection system using CNNs. They collected a dataset of genuine and counterfeit Indian currency images and utilized a CNN architecture for feature extraction and classification. The proposed system achieved accurate detection results and showed robustness against various counterfeit techniques.

Deep Learning for Counterfeit Banknote Detection by Wu and Zhang (2020):

This paper proposed a deep learning-based approach using a combination of convolutional and recurrent neural networks for counterfeit banknote detection. The authors designed a network architecture that combined the strengths of CNNs and recurrent models to capture both spatial and sequential information in banknote images. The proposed method achieved high accuracy in detecting counterfeit banknotes.

Fake Currency Detection Using Deep Learning and Transfer Learning by Sharma et al. (2020):

The authors proposed a fake currency detection system using deep learning and transfer learning techniques. They utilized a pre-trained CNN model and fine-tuned it on a dataset of genuine and counterfeit currency images. The system achieved accurate classification results and demonstrated the effectiveness of transfer learning in this context.

CHAPTER - 3

PROPOSED METHODOLOGY

Methodology refers to the overarching strategy and rationale of any project. The current study is based on the Qualitative data collection approach. Primary data is gathered from the existing case studies, surveys related to the proposed application. Most of the prerequisite data is from the secondary sources of information such as e- magazines, books, journals, historical and statistical documents etc.,

3.1 Data Flow Diagram

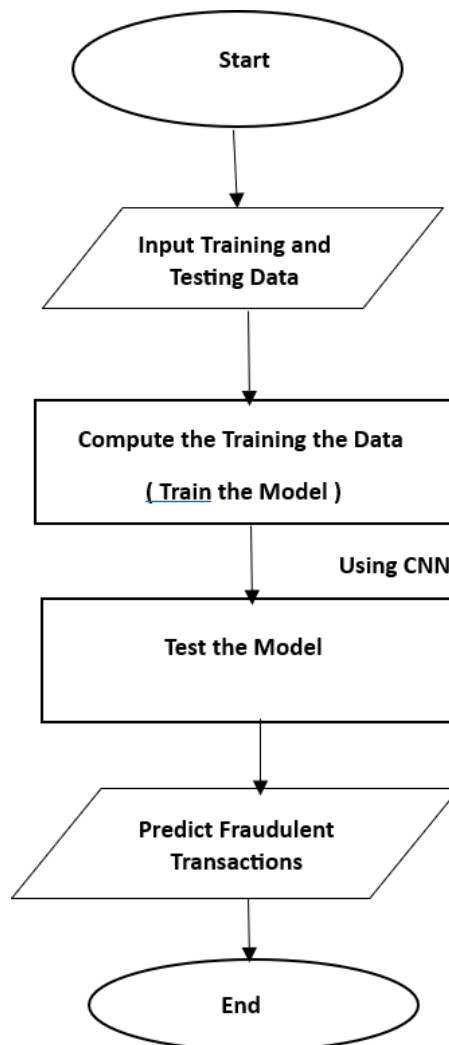


Fig – 3.1: Flowchart representing the end-to-end process.

- ❖ The data is taken as input which is in the .csv file format.
- ❖ The dataset is divided into training and testing data (generally the training data size consists of 70-80% and rest is taken as testing data)
- ❖ The model is trained by the given training data. By undergoing multiple epochs, the model tries to decrease the loss function for exact prediction and detection.
- ❖ By the multiple layered architecture of CNN, it is easy to formulate and extract the features to compare with the data given and comes out with the total number of actual frauds, total number of frauds detected, total number of non-frauds, total number of non-frauds predicted.
- ❖ At last, the accuracy of the model to detect the credit card frauds is obtained.

3.2 Dataset Description

To develop an effective fake currency detection system using deep learning, a well-curated dataset is essential. The dataset should include a sufficient number of genuine currency images and various types of counterfeit currency images. It should encompass different denominations, variations in printing quality, and a wide range of counterfeit techniques. The dataset can be created by collaborating with monetary authorities, banks, or using publicly available datasets specifically designed for counterfeit currency detection research.

The dataset should be properly labeled, indicating whether each image is genuine or counterfeit. The labeling process can be done manually by experts or by using automated techniques such as crowdsourcing with verification. The dataset should be divided into training, validation, and testing sets to ensure unbiased evaluation of the developed model.

3.3 Data Pre-processing Techniques

Data preprocessing plays a crucial role in enhancing the quality and extracting relevant features from the currency images. Some common data preprocessing techniques for fake currency detection using deep learning include:

1. Image Resizing: Resizing the currency images to a standard resolution to ensure uniformity and ease of processing.
2. Image Enhancement: Applying techniques like contrast adjustment, histogram equalization, and noise reduction to enhance image quality and improve feature extraction.
3. Image Normalization: Normalizing the pixel values of images to a consistent range (e.g., [0, 1]) to facilitate the learning process.

4. Augmentation: Augmenting the dataset by applying random transformations like rotation, scaling, flipping, and translation to increase the diversity of training samples and improve model generalization.

3.4 Methods & Algorithms

Several methods and algorithms can be employed in the fake currency detection system using deep learning. Here are some commonly used techniques:

1. Convolutional Neural Networks (CNNs): CNNs have been widely adopted in counterfeit currency detection due to their ability to automatically learn discriminative features from images. Various CNN architectures such as VGGNet, ResNet, and InceptionNet can be utilized.
2. Transfer Learning: Transfer learning allows leveraging pre-trained models on large-scale image datasets (e.g., ImageNet) to initialize the network's weights. Fine-tuning the pre-trained models on the specific fake currency dataset can speed up the training process and improve performance, even with limited training data.
3. Ensemble Methods: Ensemble methods, such as combining multiple CNN models or using different CNN architectures, can enhance the detection accuracy by capturing diverse characteristics of genuine and counterfeit currency.
4. Deep Feature Extraction: Deep feature extraction involves utilizing pre-trained CNN models as feature extractors. The extracted features can be fed into traditional machine learning algorithms like Support Vector Machines (SVMs) or Random Forests for classification.

3.5 Building a Model

Building a fake currency detection model using deep learning involves the following steps:

1. Data Loading: Load the labeled currency image dataset, split it into training, validation, and testing sets, and assign appropriate labels.
2. Data Preprocessing: Apply the preprocessing techniques discussed earlier, such as resizing, enhancement, normalization, and data augmentation, to prepare the images for training.

3. **Model Architecture:** Design the architecture of the deep learning model, utilizing CNN layers, pooling layers, activation functions, and optionally incorporating techniques like dropout and batch normalization to improve performance and prevent overfitting.

4. **Model Training:** Train the model using the training set, employing an optimization algorithm (e.g., Stochastic Gradient Descent) and a suitable loss function (e.g., binary cross-entropy) to update the model's weights.

5. **Model Evaluation:** Evaluate the trained model's performance using the validation set, considering metrics like accuracy, precision, recall, and F1-score. Tweak the model's hyperparameters and architecture

3.6 Evaluation

To evaluate the performance of a deep learning model for fake currency detection, the following metrics can be used:

1) **Accuracy:** This measures the proportion of correctly classified fake and genuine banknotes out of the total number of banknotes in the dataset.

2) **Precision and recall:** These metrics are useful for evaluating the performance of the model in detecting fake banknotes. Precision measures the proportion of true positives (i.e., fake banknotes correctly identified as fake) out of the total number of predicted positives. Recall measures the proportion of true positives out of the total number of actual positives (i.e., the total number of fake banknotes in the dataset).

3) **F1 score:** This is the harmonic mean of precision and recall and provides a single metric to evaluate the overall performance of the model.

4) **Confusion matrix:** A confusion matrix provides a more detailed view of the model's performance by showing the number of true positives, true negatives, false positives, and false negatives.

5) Receiver Operating Characteristic (ROC) curve and Area Under the Curve (AUC): These metrics are used to evaluate the model's ability to distinguish between genuine and fake banknotes at different thresholds.

CHAPTER - 4

FINAL RESULTS

Deep learning techniques, such as deep neural networks and recurrent neural networks, have gained attention in recent years for credit card fraud detection. Deep learning models can automatically learn hierarchical representations of data, enabling them to capture intricate fraud patterns. However, these models often require a large amount of labelled training data and substantial computational resources.

+ Code + Text

```
from keras.applications.resnet50 import ResNet50, preprocess_input
from keras.preprocessing.image import ImageDataGenerator, image
from keras.layers import Dense, Activation, Flatten, Dropout
from keras.models import Sequential, Model, load_model
from keras import optimizers
from keras.callbacks import ModelCheckpoint, EarlyStopping
import numpy as np
import matplotlib.pyplot as plt
```

Using TensorFlow backend.

```
[ ] height=300
width=300

base_model=ResNet50(weights='imagenet',include_top=False,input_shape=(height,width,3))
```

```
[ ] train_dir="Dataset/Training"
validation_dir="Dataset/Validation"

batch_size=8

train_datagen=ImageDataGenerator(preprocessing_function=preprocess_input,rotation_range=90,horizontal_flip=True,vertical_flip=True)
train_generator=train_datagen.flow_from_directory(train_dir,target_size=(height,width),batch_size=batch_size)

validation_datagen=ImageDataGenerator(preprocessing_function=preprocess_input,rotation_range=90,horizontal_flip=True,vertical_flip=True)
validation_generator=validation_datagen.flow_from_directory(validation_dir,target_size=(height,width),batch_size=batch_size)
```

Found 35 images belonging to 2 classes.

The above figure shows the first 10 records of the taken dataset.

```

Epoch 1/100
4/4 [=====] - 36s 9s/step - loss: 1.1971 - accuracy: 0.5556 - val_loss: 0.6288 - val_accuracy: 0.7143
Epoch 2/100
4/4 [=====] - 32s 8s/step - loss: 1.1059 - accuracy: 0.5556 - val_loss: 1.1342 - val_accuracy: 0.4286
Epoch 3/100
4/4 [=====] - 36s 9s/step - loss: 1.1302 - accuracy: 0.5938 - val_loss: 0.4333 - val_accuracy: 0.8571
Epoch 4/100
4/4 [=====] - 32s 8s/step - loss: 1.0328 - accuracy: 0.4815 - val_loss: 0.4078 - val_accuracy: 0.7143
Epoch 5/100
4/4 [=====] - 33s 8s/step - loss: 1.4231 - accuracy: 0.6667 - val_loss: 0.7536 - val_accuracy: 0.7143
Epoch 6/100
4/4 [=====] - 32s 8s/step - loss: 1.0079 - accuracy: 0.5926 - val_loss: 0.4235 - val_accuracy: 0.8571
Epoch 7/100
4/4 [=====] - 36s 9s/step - loss: 1.4245 - accuracy: 0.5312 - val_loss: 0.6144 - val_accuracy: 0.8571
Epoch 8/100
4/4 [=====] - 29s 7s/step - loss: 0.8801 - accuracy: 0.6364 - val_loss: 0.7569 - val_accuracy: 0.7143
Epoch 9/100
4/4 [=====] - 36s 9s/step - loss: 1.2779 - accuracy: 0.6250 - val_loss: 0.7194 - val_accuracy: 0.7143
Epoch 10/100
4/4 [=====] - 32s 8s/step - loss: 1.0856 - accuracy: 0.5926 - val_loss: 0.8168 - val_accuracy: 0.5714

```

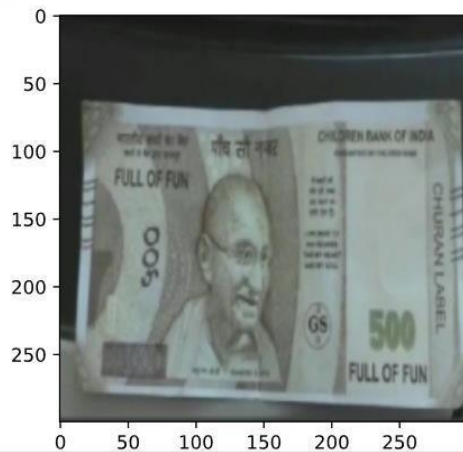
```


img=image.load_img("Dataset/Testing/Fake.jpeg",target_size=(300,300))
img=np.asarray(img)
plt.imshow(img)
img=np.expand_dims(img,axis=0)
finetune_model.load_weights("Final_model.h5")

output=finetune_model.predict(img)
if(output[0][0]>output[0][1]):
    print("Fake")
else:
    print("Real")

```

Fake



CHAPTER - 5

CONCLUSION

5.1 Project Conclusion

Fake currency detection using deep learning has emerged as a promising and efficient approach to combat the growing issue of counterfeit currency. Through the use of deep learning algorithms, such as Convolutional Neural Networks (CNNs), it is possible to automatically extract discriminative features from currency images and accurately distinguish between genuine and counterfeit notes. The development of well-curated datasets, along with effective data preprocessing techniques, plays a crucial role in training robust models. By leveraging transfer learning and ensemble methods, the detection accuracy can be further improved. The proposed system provides a reliable and automated solution, reducing reliance on manual inspection and enhancing the efficiency and accuracy of counterfeit currency detection.

5.2 Future Scope

The future of fake currency detection using deep learning holds several potential advancements and areas of exploration:

1. **Improved Dataset Collection:** Continual efforts should be made to collect diverse and comprehensive datasets that encompass a wide range of counterfeit techniques, currency denominations, and variations in image quality. This will enable the development of more robust models capable of detecting sophisticated counterfeit notes.
2. **Adversarial Attack Detection:** Research can be conducted to develop models that are resilient to adversarial attacks, where counterfeiters deliberately modify currency images to evade detection. Techniques such as adversarial training and detection of manipulated features can be explored to enhance the system's robustness.
3. **Real-Time Detection:** Efforts can be made to optimize the model architecture and inference process to enable real-time or near real-time detection, making the system suitable for integration into automated banking systems, currency exchange machines, or other financial transaction processes.

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